

INSTALLATION GUIDE | Ubuntu Server and Windows Server Deployment Using VirtualBox

ITEP 414 – System Administration and Maintenance

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Project Overview

Operating Systems serve as the foundation of every enterprise IT infrastructure. Before a server can host websites, databases, cloud services, or enterprise applications, it must first be properly installed, configured, secured, and documented.

In this project, you will assume the role of a Junior System Administrator responsible for deploying a new Ubuntu Server for a startup company. You will perform a complete operating system installation, configure essential settings, verify the installation, compare modern boot technologies, and create professional documentation suitable for future administrators. The deliverables from this project will become part of your professional GitHub portfolio

Objectives

After completing this project, students should be able to:

Knowledge

Students should be able to:

1. Explain the purpose of an operating system in enterprise environments.
2. Differentiate BIOS and UEFI firmware.
3. Explain the stages of the computer boot process.
4. Compare Ubuntu Server, Windows Server, and Rocky Linux.

Skills

Students should be able to:

- Install Ubuntu Server in a virtual machine.
- Configure server settings during installation.
- Enable secure remote administration using SSH.
- Verify server functionality.
- Document installation procedures.
- Produce professional technical documentation.

Professional Outcomes

By the end of this activity, students will have:

- Installed Ubuntu Server
- Configured a working Linux server
- Produced a deployment guide
- Created installation documentation
- Updated their GitHub portfolio

- Published a LinkedIn learning reflection

Software Requirements

Software	Version	Purpose
Oracle VirtualBox	Latest Stable Version	Virtualization platform
Ubuntu Server	24.04 LTS	Linux server operating system
Windows Server Evaluation	2022 Standard Evaluation	Windows server operating system
Draw.io / diagrams.net	Latest Version	Boot process flowchart creation
Microsoft Word	2019 or later	Documentation preparation

Virtual Machine Specifications

Component	Ubuntu Server VM	Windows Server VM
VM Name	Ubuntu-Server-Week03	Windows-Server-Week03
RAM	4 GB	4 GB
CPU	2 Virtual Processors	2 Virtual Processors
Storage	40 GB VDI	50 GB VDI
Network	NAT	NAT
Optical Drive	Ubuntu Server ISO	Windows Server 2022 Evaluation ISO

Step-by-Step Installation Procedure

Step 1 – Create the Virtual Machine

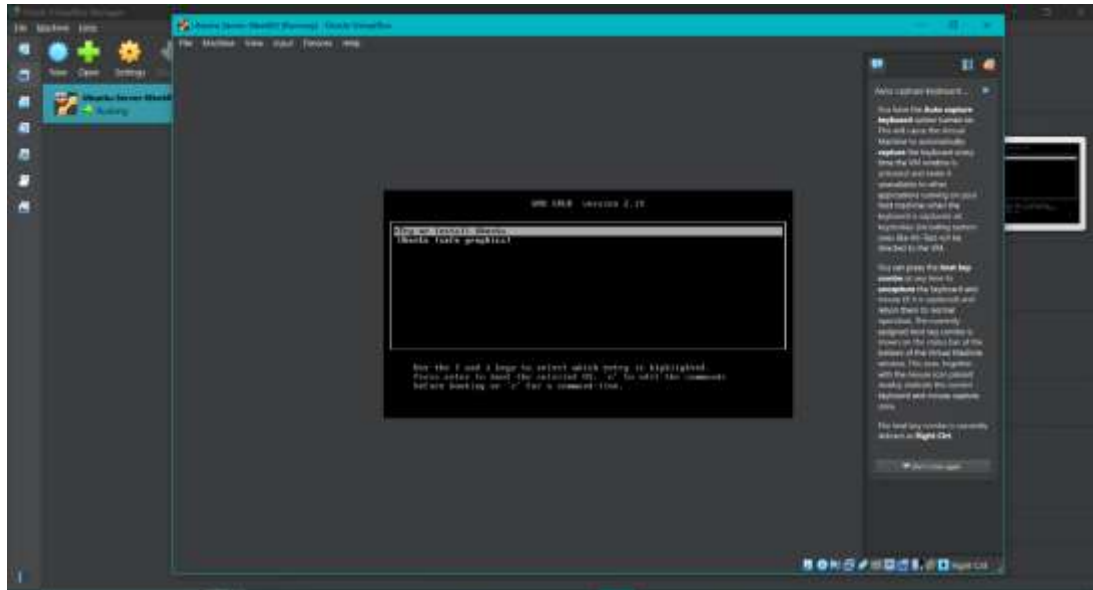
1. Open Oracle VirtualBox.
2. Click New.
3. Configure the VM using the specifications above.

Step 2 – Attach the Ubuntu ISO

1. Open Settings → Storage.
2. Select the Empty optical drive.
3. Attach the Ubuntu Server ISO.

Step 3 – Start Installation

1. Click Start.
2. Choose Try or Install Ubuntu Server.
3. Select Install Ubuntu Server.



Step 4 – Configure Installation Settings

- Language: English
- Keyboard: English (US)
- Network: DHCP
- Hostname: server01
- Username: sydneycoleen
- Disk Setup: Guided – Use Entire Disk
- File System: ext4
- Install OpenSSH Server: Enabled

Step 5 – Complete Installation

- Wait for the installation to finish.
- Click Restart Now.
- Remove the installation ISO when prompted.

Configuration Details

Setting	Value
Hostname	server01
Username	sydneycoleen
File System	ext4
Network Mode	NAT
SSH Server	Enabled
Package Manager	APT

Verification Commands

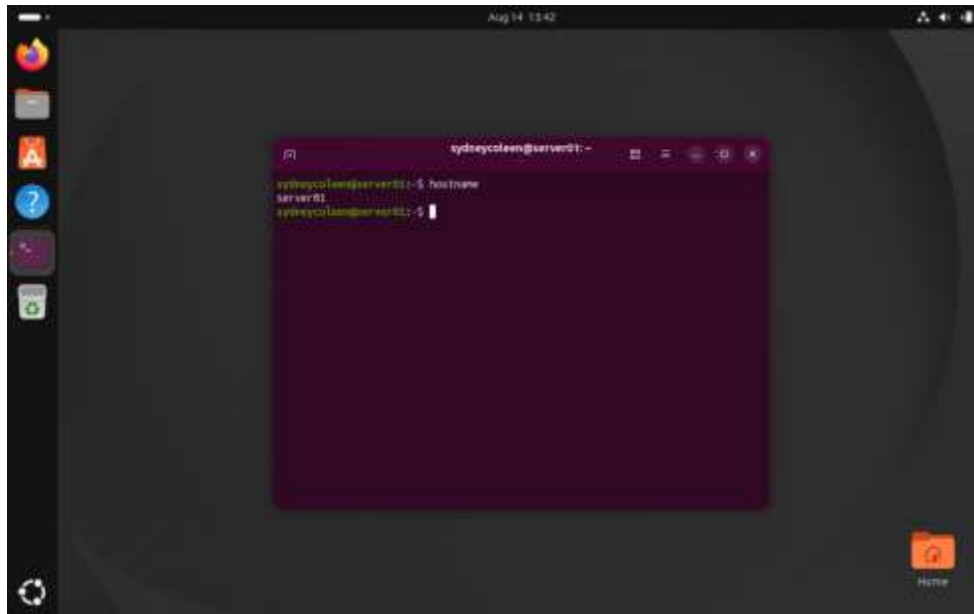
After logging in, the following commands were executed.

Verify Hostname

hostname

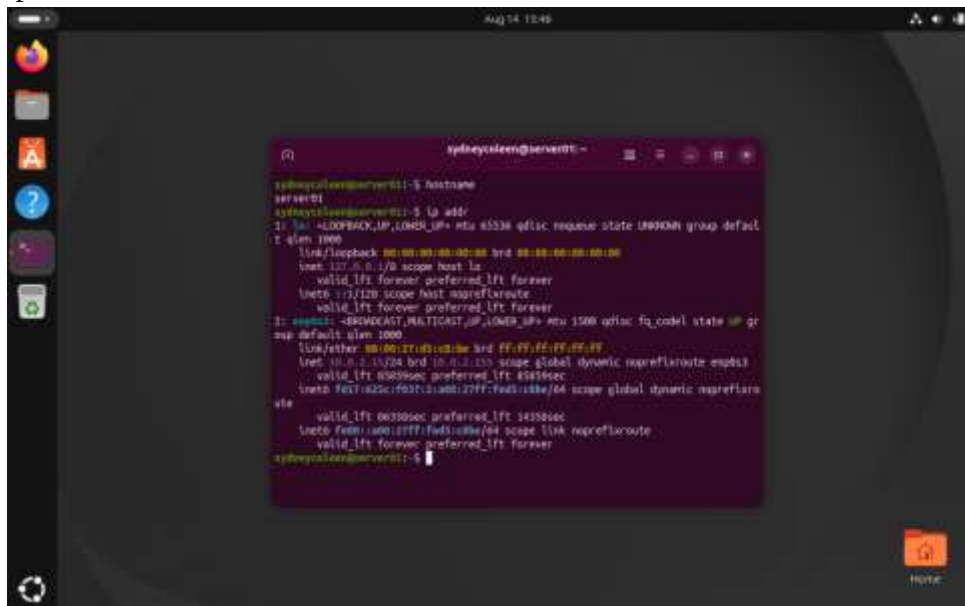
Expected Output:

server01



Verify IP Address

ip addr



```
ping -c 4 google.com
```

The screenshot shows a Linux desktop with a dark theme. On the left is a vertical dock with icons for the Dash, Firefox, Files, App Center, a question mark, a terminal, and a green icon. The top status bar shows the date and time as 'Aug 14, 12:34'. The main window is a terminal titled 'xylapcsleev@server01:~' with a dark purple background. The terminal output shows the results of a ping command to google.com, including packet statistics and RTT values.

```
xylapcsleev@server01:~$ ping -c 4 google.com
PING google.com (172.217.26.78) 56(84) bytes of data:
64 bytes from sl12b02-ls-f76.3a88.net (172.217.26.78): icmp_seq=1 ttl=64 time=
8.69 ms
64 bytes from sl12b02-ls-f76.3a88.net (172.217.26.78): icmp_seq=2 ttl=64 time=
8.74 ms
64 bytes from sl12b02-ls-f76.3a88.net (172.217.26.78): icmp_seq=3 ttl=64 time=
11.3 ms
64 bytes from sl12b02-ls-f76.3a88.net (172.217.26.78): icmp_seq=4 ttl=64 time=
8.68 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 380ms
rtt min/avg/max/mdev = 8.68/8.55/11.94/73.379 ms
xylapcsleev@server01:~$
```

```
sudo apt update
```

[illegible]

[illegible]

Active: active (running)

```

systemctl status ssh

Created symlink '/etc/systemd/system/ssh.service' → '/usr/lib/systemd/system/ssh.service'.
Created symlink '/etc/systemd/system/multi-user.target.wants/ssh.service' → '/usr/lib/systemd/system/ssh.service'.
systemd[1]: ssh.service:
Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled); preset: enabled
Active: active (running) since Fri 2020-08-14 14:38:58 PDT; 5ss ago
Invocation: 45aa80e10eeef0d94c32d40bec0133a
TriggeredBy: ● ssh.socket
Doc: man:ssh(8)
Main PID: 17417 (sshd)
Tasks: 2 (limit: 4096)
Memory: 1.2M (peak: 2.1K)
CPU: 3ms
CGroup: /system.slice/ssh.service
        └─17417 "sshd: /usr/sbin/sshd -D [listens] * of 20 ssh daemons"

Aug 14 14:38:58 server01 systemd[1]: Starting ssh.service - OpenSSH Secure Shell.
Aug 14 14:38:58 server01 sshd[17417]: Server listening on 0.0.0.0 port 22.
Aug 14 14:38:58 server01 sshd[17417]: Server listening on :: port 22.
Aug 14 14:38:58 server01 systemd[1]: Started ssh.service - OpenSSH Secure Shell.

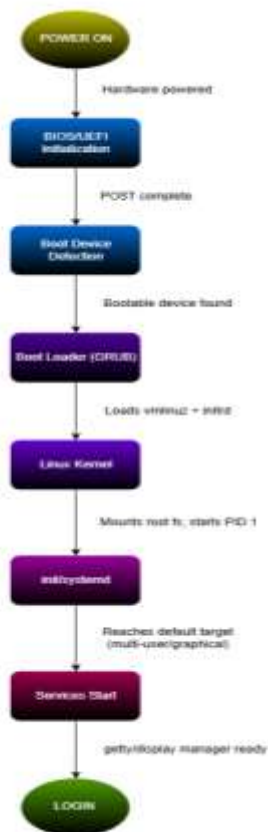
```


BIOS vs UEFI Comparison

(Detailed Comparison is located at Week 3 Repository, named BIOS_vs_UEFI_&_OS_Comparison_Report.pdf)

Feature	BIOS	UEFI
Definition	Traditional firmware interface	Modern firmware interface
Boot Process	16-bit, MBR-based	32/64-bit, GPT-based
Maximum Disk Support	Up to 2 TB	More than 2 TB
Partition Style	MBR	GPT
Security Features	Limited	Secure Boot support
Speed	Slower	Faster
Advantages	Simple and compatible with older hardware	Faster, secure, supports large disks
Disadvantages	Limited features and disk support	More complex configuration
Modern Usage	Legacy systems	Modern computers and servers

Boot Process Flowchart



Windows Server Installation Summary

(Detailed Installation is located at Week 3 Repository, named BIOS_vs_UEFI_&_OS_Comparison_Report.pdf)

Installation Procedure

1. Create a new VM.
2. Attach the Windows Server 2022 Evaluation ISO.
3. Select Custom: Install Windows only (advanced).
4. Choose Windows Server 2022 Standard Evaluation (Desktop Experience).
5. Create the Administrator password.
6. Log in successfully.
7. Rename the computer to SERVER01-WS.

Configuration Summary

Setting	Value
Operating System	Windows Server 2022 Standard Evaluation
Installation Type	Desktop Experience
Computer Name	SERVER01-WS
Administrator Account	Administrator

OS Comparison Table

(Detailed Comparison is located at Week 3 Repository, named BIOS_vs_UEFI_&_OS_Comparison_Report.pdf)

Feature	Windows Server	Ubuntu Server	Rocky Linux
Licensing	Commercial / Proprietary	Free and Open Source	Free and Open Source
User Interface	GUI and Server Core	Command Line (optional GUI)	Command Line (optional GUI)
Package Management	Windows Update, MSI, PowerShell, Winget	apt / dpkg	dnf / rpm
Security	Active Directory, Group Policy, Defender	AppArmor, UFW	SELinux, firewallld
Performance	Higher resource usage	Lightweight	Stable and optimized
Enterprise Use Cases	AD, file servers, Microsoft applications	Web servers, cloud, containers	Enterprise Linux, databases, virtualization

Advantages	Easy GUI administration	Free, flexible, cloud-friendly	Stable, RHEL-compatible
Disadvantages	Licensing cost, resource intensive	Steeper learning curve for GUI-focused users	Smaller community and fewer beginner-oriented resources

Problems Encountered

Problem	Description
Ubuntu package update failed	Failed to fetch and 403 Forbidden errors occurred during apt update.
DNS resolution issue	google.com: Name or service not known was encountered.
SSH service missing	ssh.service could not be found after installation.

Solutions Implemented

Problem	Solution
Package update failed	Changed the Ubuntu mirror from ph.archive.ubuntu.com to archive.ubuntu.com.
DNS issue	Verified NAT networking and tested connectivity using ping 8.8.8.8.
SSH missing	Installed OpenSSH Server using sudo apt install openssh-server -y.

Lessons Learned

Through this laboratory activity, I learned how to deploy both Linux and Windows server operating systems in a virtualized environment. I became familiar with creating virtual machines, configuring storage and networking, and performing post-installation verification tasks. I also gained practical experience in troubleshooting package repository errors, DNS connectivity issues, and missing services in Ubuntu Server.

The activity improved my understanding of the differences between BIOS and UEFI, particularly regarding boot mechanisms, partition styles, security features, and modern hardware support. Creating the Ubuntu boot process flowchart helped me understand how firmware, GRUB, the Linux kernel, and systemd interact during system startup.

Comparing Windows Server, Ubuntu Server, and Rocky Linux showed that each operating system has strengths depending on enterprise requirements. Windows Server is well suited for Microsoft-centered infrastructures, Ubuntu Server excels in cloud and web environments, and Rocky Linux provides enterprise-grade stability and compatibility with Red Hat Enterprise Linux.

Overall, this project strengthened my skills in system administration, virtualization, server configuration, troubleshooting, and technical documentation, which are essential competencies for a future IT professional.

References

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